

NITISH KUMAR SAH

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Work Experience

Saptang Labs – Computer & Network Security

Dec 2025 – Present

Software Engineer Intern

Remote

- Designed and developed **production-grade backend REST APIs** using FastAPI, delivering **25+ secure endpoints** with **role-based access control (RBAC)** and authentication workflows for user and client management.
- Engineered **activity logging and audit pipelines** capturing **100% critical user actions**, enabling compliance-ready monitoring and forensic traceability.
- Implemented **pagination and query optimization** techniques, reducing average API response time by **30–40%** and improving database efficiency for high-volume requests.

Projects

CampusSentinel – AI Campus Surveillance (Runners-Up at IIT Guwahati)

[GitHub](#) - [Demo](#)

Django REST Framework, FastAPI, PostgreSQL (pgvector), LangGraph, JWT

- Designed a **relational PostgreSQL data model** with **12+ indexed tables** to correlate multi-source campus events.
- Built a **microservice backend** (Django REST Framework+FastAPI) with **10+ endpoints** and JWT authentication.
- Architected the **A.R.I.A. FastAPI agent** with **four tool integrations**, enabling orchestration of API workflows.

MediMind – AI-Powered Medical Assistant Platform

[GitHub](#) - [Live](#)

Django REST Framework, FastAPI, PostgreSQL, LangChain, Google Gemini 2.5, Pydantic

- Built **REST APIs (15+ endpoints)** with JWT and a normalized **PostgreSQL schema** deployed on Supabase.
- Deployed **Django REST backend** with Gunicorn and a **FastAPI AI microservice** with Uvicorn on Render.
- Integrated **LangChain + Google Gemini 2.5** in FastAPI to generate medical outputs with Pydantic v2 validation.

Technical Skills

- Languages:** Python, C++, JavaScript, SQL
- Databases:** MongoDB, PostgreSQL (pgvector), MySQL, Pinecone, Redis
- Frameworks & Libraries:** FastAPI, Django REST Framework, React.js, LangChain, Pydantic, JWT
- Cloud & DevOps:** AWS (EC2, RDS), Docker, Git, GitHub, Gitea, Cloudinary
- Tools:** VS Code, PyCharm, MongoDB Atlas, WebStorm, PgAdmin4, Anaconda

Education

National Institute of Technology, Rourkela

Aug 2023 – Present

B.Tech in Biotechnology

Rourkela, Odisha

DAV Public School, Hehal, Ranchi

2022

CBSE, Science (PCM)

Ranchi, Jharkhand

Achievements/Certifications

- Runners-Up, Ethos 2025, IIT Guwahati** [Certificate](#)
- 2nd Runners-Up, Hackfest'25, IIT-ISM Dhanbad** [Certificate](#)
- Supervised Machine Learning: Regression and Classification – DeepLearning.AI** [Certificate](#)
- Advanced Learning Algorithms – DeepLearning.AI** [Certificate](#)

Extracurricular Activities

Pantomime - Dramatics Club

Mar 2024 – Present

Treasurer

NIT Rourkela

Webwiz - Web Development Club

Jan 2025 – Present

Technical Team Member

NIT Rourkela

Innovision - Technical Fest

Sep 2024 – Nov 2024

Coordinator

NIT Rourkela